services. There are a lot of combinational architectures that Google and Azure are coming out with (both edge and hybrid).

Today, data storage is not a serious problem as gigabytes and terabytes space is available pretty much on every device. The battery life is also gradually improving in terms of longevity. But the biggest challenge would be to have optimised or purpose-built algorithms for edge devices, IoT devices, gateways etc.

This is where innovations are taking place. On the hardware level as an example, Google's AI-edge stack has an edge TPU, called CoralEdge, and an IoT edge software pack that brings in the entire Tensor framework algorithmic pieces as lightweight into mobile devices.

The iPhone 12 has A14 chip that is purpose-built for AI and ML. Called Deep Fusion, it is hybrid and has 8-core neural engines, 6x faster matrix multiplication, one trillion hours per second on CPU, and 40 per cent more power efficiency, making the compute cycle and instruction execution efficient. NVIDIA is also not far behind. Facebook has PyTorch mobile. In terms of difference, each is optimised for battery, storage, and other device needs.

The currently released GPUs are basically vector processing units built into the SoC itself. It is like integrating GPU inside the CPU, and giving a combined solution on the edge, which can do all of those calculations really fast.

Solving the latency issue

For the data, which is being produced at millisecond-level or even second-level, you need some kind of analytics on the edge so that you can prioritise and make immediate decisions, and choose to store aggregated data.

There are two dimensions to latency: propagation delay (the time taken for data to go and come back, which is decided by your IP routes and routing cables) and the time taken to run compute-intensive AI and ML algorithms like CNNs and neural networks. So, if you are running resources in a high-computing data centre with mainframes, then your time would be relatively less compared to a normal phone, which has a limited MIPS capability.

Companies like NVIDIA, Google, and even Apple have purpose-built AI-chips for running heavy computational loads with respect to AI and ML. Google's TFUs and TPUs run at a much faster rate on par with mainframes and big machines. These tools are coming to the edge today as there is a need and value for businesses and people. It is essential to have hardware with good network and compute speed along with the technical stack such as TensorFlow.

Whenever designing or developing an AI or ML system, you should take into consideration the required response speed along with the expectation of the user. Depending on that, you will have to tweak your algorithms and data flow.

The applications aka use-cases

From the consumer side, there are many applications of doing AI or ML on edge—from voice assistant, face recognition to wearables. From a B2B perspective, it is Industrial IoT (IIoT).

"A French company, which manufactures components for aircraft, uses real-time location solution (RTLS) like RFID or BLE to track different parts. Since new parts keep coming in when old parts are consumed, the company was not able to understand the usage frequency of a particular component in a manufacturing line. Therefore they made gateways, which were deployed in the warehouse and started implementing ML on them. The gateway was power over ethernet (PoE), which has a good processor capability. Through this, they were able to process the information locally and the operator on the floor got quickly informed about the outcome. This was done through multiple ML algorithms on data over a period of one year. This was not done real-time but based on predictions made by the algorithm. Ninety per cent of the time, the predictions came out to be true. If it was done real-time, it would have taken a lot of time to make a decision and give the output to the operator," informs Vinay Solanki.

He further adds that his company, Napino, leverages AI on edge as robots are used for evaluation purposes. An in-house solution is also being built for social distancing and marking attendance of employees using IoT applications with beacons. This is not yet an AI implementation, but the application is going to have a lot of data on which ML will be implemented to find analytical insights. Another application that was worked upon was a smart shower, which monitored the amount of water consumed while taking a bath. That was achieved using a flow meter plus an IoT gateway to upload the data.

According to Dharmendra Kumar, a major application for AI on edge is industry asset monitoring. Machines need to be controlled and monitored for optimal data consumption and processing. If this happens on AI-edge, then data acquisition will be fast and happen inside edge only rather than data going to the cloud and then giving output. On the consumer side, home automation seems to be big.

Venkateswar Melachervu states a very unique use-case. A year or two ago, he was working on providing a telemedicine platform for pregnant women—right from conception to postpartum. Since a lot of doctor consultations and tests take place over a period of nine months, the idea was to offer these services at convenience with the help of an app. This was a noble step in timely addressing certain critical diseases that can manifest during the pregnancy such as gestational diabetes mellitus, which is characterised by sugar levels becoming high during pregnancy, affecting the health of both the baby and the mother.

Because of the computational intensity and data, it was chosen to run a model in the cloud, get the data and give the results. But with AI-edge coming in, a model can now be created on the edge itself, based on the data of 10,000 to 20,000 pregnant women. With sufficient data, it is possible to derive a linear prediction model to arrive at a risk percentage (between 0 and 1). It can help in predicting early-on gestational diabetes with the help of ML, which can be constantly updated as the results keep coming in.

He also states another use-case. Electricity meter reading is generally